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AI Infrastructure & Systems Engineer · 3rd Year CSE (AI & ML)
Go · Rust · TypeScript · Python

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SUMMARY

I'm a 3rd-year CSE (AI & ML) student who ships **infrastructure, not demos**. I built **AgentMesh** — a framework-agnostic AI-agent control plane — across all 8 planned milestones, and **CodeGraph**, a live 7-agent code-remediation swarm passing 96 of 96 tests with *zero* LLM API dependency. Determinism isn't a value statement for me; it's a habit I built after getting burned by its absence. I self-audited CodeGraph at 3/10, published it openly, then followed up with a 99-issue security review and fixed what it found. Nothing ships until I can explain *why* it's correct — not just that it worked once.

8/8 AGENTMESH MILESTONES	96/96 CODEGRAPH TESTS	6 PROJECTS SHIPPED	4 LANGUAGES IN PRODUCTION
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SELECTED WORK

AgentMesh 2026 · Shipped · github.com/archdex-art/AgentMesh
Framework-agnostic control plane for AI agents

- Designed a schema-as-source-of-truth data model over ClickHouse + Postgres + Redis + MinIO for full-fidelity agent telemetry.
- Built framework adapters for LangGraph, CrewAI, AutoGen, and the OpenAI Agents SDK — one OTel-native integration point.
- Implemented a **deterministic replay engine** that reproduces historical agent failures exactly, enabling postmortem debugging without reproductions.
- Shipped full MCP governance layer: policy enforcement, capability scoping, and audit trails across evolving agent frameworks.
- Completed all 8 planned milestones with CI, Helm charts, containers, and deploy pipeline included.

Go

OpenTelemetry

ClickHouse

Postgres

Redis

MCP

Distributed Systems

CodeGraph 2026 · Live on Render · github.com/archdex-art/CodeGraph
Symbol-level knowledge graph + deterministic 7-agent remediation swarm

- Turns any git repo into a traversable symbol graph; implemented Tarjan SCC + BFS analysis with blast-radius-weighted health scoring.
- Deterministic 7-agent swarm: sandbox → codemod → re-index → verified diff — no LLM API key required, 96/96 test suite green.
- Engineered a **verify-before-apply** pipeline: fixer clones to a disposable sandbox, applies codemod, re-indexes, and only emits a diff if health score doesn't regress.

- Self-audited at 3/10, published results openly, then followed up with a 99-issue security review — hardening tracked 3/10 → 8/10.
- Shipped as a hosted Next.js app (Render) and a fully standalone offline VS Code extension (CodeGraph Local).

TypeScript

Next.js

node:sqlite

Static Analysis

Multi-Agent

VS Code API

TS Compiler API

SuperSearch

2026 · Active · github.com/archdex-art/SuperSearch

Intent-driven runtime for macOS — natural language → replayable task graphs

- Compiles natural language into replayable, inspectable execution graphs via an **object-capability security model** with an append-only journal.
- Built a WASM plugin host (wasmtime) with fuel and memory limits — sandboxed third-party extensions with deterministic resource budgets.
- Capability gate + argv-only execution path enforced on every live dispatch: no shell injection, no ambient authority.
- Crafted a Spotlight-grade command palette in React + Framer Motion over Tauri IPC for a native macOS feel.

Rust

Tauri

WASM

React

Security

Framer Motion

Live API Inspector

2026 · Active · github.com/archdex-art/live-api-inspector

Privacy-first Chrome DevTools extension for network diagnostics

- Semantic endpoint clustering, performance intelligence, and local secret redaction — all analysis stays on-device.
- Zero-config DevTools install; built with TypeScript, React 19, Vite, and Manifest V3.

TypeScript

React 19

Manifest V3

Vite

BrowseMesh

2026 · Active · github.com/archdex-art/BrowseMesh

Local-first browser knowledge graph with hybrid search

- Entity extraction, embedding pipeline, and hybrid full-text + vector search over everything you've read — offline, private, no server.
- Plugin SDK for extensibility; all storage via IndexedDB behind a typed data tier.

TypeScript

Vector Search

IndexedDB

Manifest V3

TECHNICAL SKILLS

LANGUAGES

Go

Rust

TypeScript

Python

AI & AGENT INFRASTRUCTURE

OpenTelemetry

Model Context Protocol

LangGraph

CrewAI

AutoGen

OpenAI Agents SDK

Vector Search

Deterministic Replay

SYSTEMS & BACKEND

ClickHouse

Postgres

Redis

MinIO

gRPC / Protobuf

WASM (wasmtime)

Tauri

Docker

Helm

FRONTEND & DEVELOPER TOOLING

Next.js

React

Framer Motion

Tailwind CSS

VS Code API

TS Compiler API

Manifest V3

Vite

SECURITY & CORRECTNESS

- Object-Capability Model
- Append-Only Journals
- Sandbox Execution
- Static Analysis
- Adversarial Testing
- Open Security Audits

ENGINEERING PHILOSOPHY

P01 · DETERMINISM OVER MAGIC

Same input, same result — every run. CodeGraph's swarm operates without an LLM key; SuperSearch compiles intent into inspectable, replayable graphs.

P02 · SECURITY IN THE OPEN

Self-audits get published, not buried. CodeGraph's posture was tracked from 3/10 → 8/10 openly, with a 99-issue postmortem shipped in the clear.

P03 · VERIFY BEFORE APPLY

The fixer sandboxes a fix, applies it, re-indexes, and only emits a diff if the health score doesn't regress. Sandbox → codemod → re-index → diff.

P04 · SHIP THE WHOLE THING

Research isn't done until it deploys. Every project carries its own CI, docs, containers, and deploy path — including the parts that didn't work the first time.

EDUCATION

Keshav Memorial College of Engineering
B.Tech — Computer Science Engineering (AI & ML)
August 2024 – 2028 · Hyderabad, Telangana